

Key Takeaways

What Metric3D Got Right

- CSTM identified the fundamental problem: focal length creates metric ambiguity in mixed-camera training
- Joint depth-normal optimization with consistency loss was novel and effective
- Proved that large-scale metric depth training is possible

Where the Field Has Moved

- Camera intrinsics should be learned, not required as input (Depth Pro, UniDepth)
- Self-supervised backbones (DINOv2) outperform supervised ViT for dense prediction
- Synthetic teacher + pseudo-label pipeline (Depth Anything v2) beats curated labeled data

Open Questions

- Is metric depth from a single image the right abstraction, or should we learn camera-aware representations?
- Can video depth models replace per-frame methods entirely?
- How to handle domains with no training data (underwater, medical, aerial)?